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1. A brief introduction of yourself and your interests.

Hello! I am Sumedha, currently in the second semester of my Master's in Data Analytics and Computational Social Science (DACSS). I have a keen interest in applying computational techniques to real-world problems, particularly in the field of Politics, Government Policies, Women Empowerment, and Education. My academic journey has equipped me with a solid foundation in statistics and data analysis. With this course, I am excited to learn and utilize NLP to extract meaningful insights from large textual datasets.

My hobbies include cooking and shopping!

2. A clear description of your research question.

Research Question - What are the prevalent patterns and key themes in consumer complaints about financial products?

With this research question, I aim to uncover common issues and sentiments expressed by consumers in their complaints regarding financial products and services. By applying NLP techniques, we can systematically analyze large volumes of textual data to identify recurring themes, sentiments, and emerging trends. This analysis will help in understanding the primary concerns of consumers and how they feel about various financial products and services.

3. A clear description of your corpus and where you plan to access it.

The primary dataset for this research is the Consumer Complaint Database maintained by the U.S. Consumer Financial Protection Bureau (CFPB). This comprehensive database contains over 1.5 million consumer complaints related to a wide range of financial products and services, including but not limited to credit cards, mortgages, loans, and debt collection. Each complaint includes a narrative provided by the consumer, detailing their specific issues and experiences.

The dataset is publicly accessible and can be downloaded in CSV format from the following sources:

- [Data.gov](https://data.gov)
- [Consumer Financial Protection Bureau Dataset](https://cfpb.com/consumer-complaint-database)

These platforms provide the most recent data, which is essential for conducting up-to-date analyses.

4. A rationale for the corpus that identifies why it is appropriate for addressing the selected research question.

The CFPB's Consumer Complaint Database is particularly well-suited for this research due to several key factors:

- **Richness of Data:** The database offers detailed consumer narratives that provide in-depth insights into individual experiences and issues with financial products.
- **Volume and Diversity:** With over 1.5 million complaints spanning various financial products and services, the dataset is large and diverse enough to support comprehensive text analysis and to identify patterns across different demographics and financial sectors.
- **Public Accessibility:** Being publicly available, the dataset allows for transparent and reproducible research.

Appropriateness for Addressing the Research Question:

Analyzing this corpus enables the application of NLP techniques to extract meaningful patterns and sentiments from the data. For instance, topic modeling can uncover prevalent themes in consumer complaints, while sentiment analysis can assess the overall emotional tone of the narratives. This approach facilitates a systematic examination of consumer issues, leading to a better understanding of common grievances and areas where financial institutions may need to improve.

Real-World Applications

The insights derived from this analysis have several practical implications:

- **For Financial Institutions:** Understanding common consumer complaints can help institutions identify areas needing improvement, enhance customer service, and develop better products tailored to consumer needs.

- For Regulators: Regulatory bodies can monitor emerging issues in the financial sector, assess the effectiveness of existing regulations, and identify areas where new policies may be required to protect consumers.
- For Consumers: Publicly available analyses can inform consumers about common issues with financial products, aiding them in making more informed decisions.

By leveraging NLP techniques to analyze consumer complaint narratives, this research aims to contribute valuable insights that can lead to improved financial products, better regulatory policies, and enhanced consumer protection.